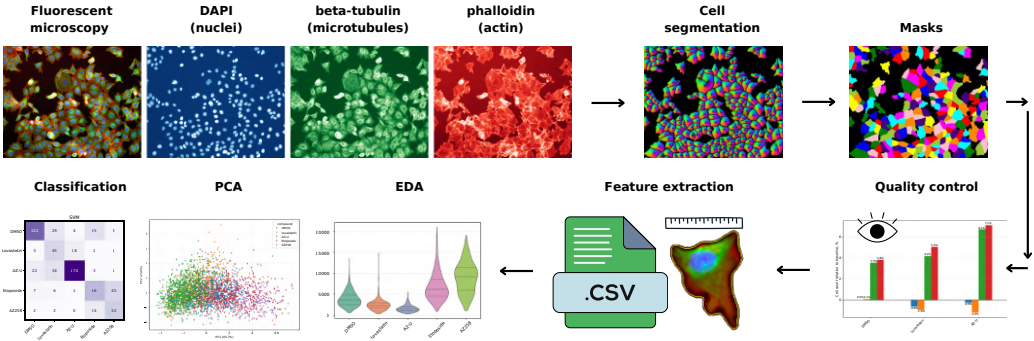
Cellpose-Based Segmentation for Extracting Morphological Features from Drug-Treated Cancer Cell Images

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Graphical abstract



The project uses a subset of BBBC021

- MCF-7 breast cancer cells, 24 h drug treatment.
- Full dataset: 13,200 fields of view; 113 compounds in 8 concentrations.
- Each field of view has 3 TIFF files: DAPI, beta-tubulin and actin.
- Project subset: 52 fields of view: 48 treatment fields of view (4 compounds $\times$ 3 concentrations $\times$ 4 fields per compound-concentration pair) plus 4 DMSO control fields.
- This gives 52 fields of view $\times$ 3 channels $=$ 156 TIFF files.

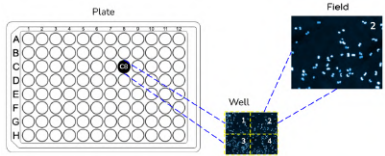


Figure 1. Plate-well-field hierarchy

Table 1. Compounds chosen

MoA group	Compound	Middle dose
control	DMSO	0.0
cholesterol-lowering	Lovastatin	5.0
epithelial	AZ-U	3.0
DNA damage	Etoposide	3.0
Aurora kinase inhibitor	AZ258	0.3

Cell appearance

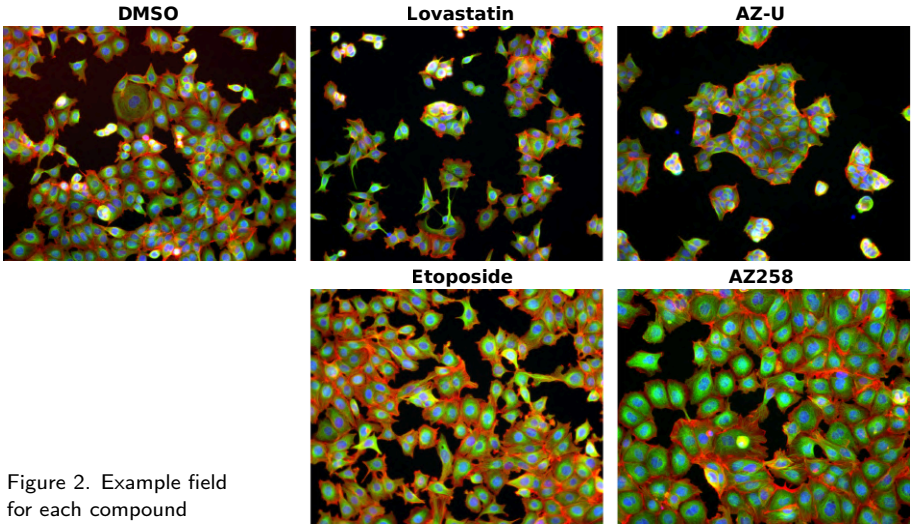


Figure 2. Example field for each compound

Watershed baseline and its Stardist-marker modification against Cellpose

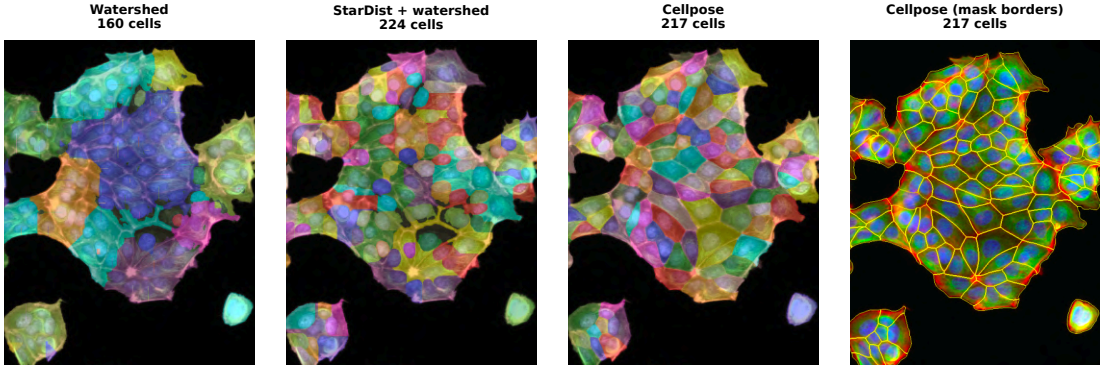


Figure 3. AZ-U masks

Note

Here and further, images are cropped and zoomed to show not a field, but a few cells for clarity

Cellpose tuning grid search

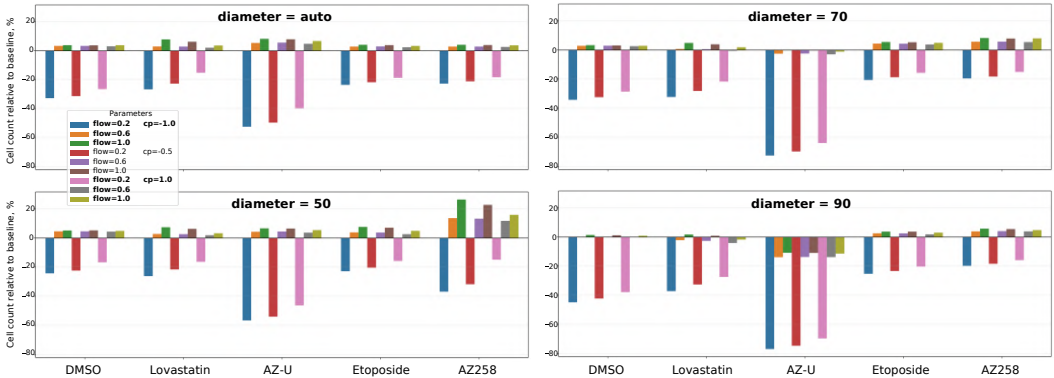


Figure 4. Cell count change relative to baseline Cellpose

Diameter	Cell probability threshold	Flow threshold
Object scale in pixels used to resize the image before segmentation	Controls which pixels are accepted as cellular foreground	Mask-quality filter based on flow consistency

Default configuration: diameter = None; cellprob_threshold = 0.0; flow_threshold = 0.4

Fixed diameter creates different errors in different phenotypes

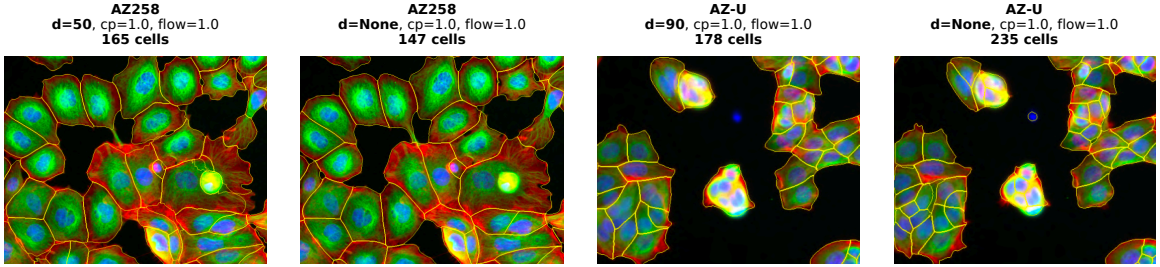


Figure 5. Cellpose diameter grid-search crops

Too small diameter

Large cells can be split into several masks or their boundaries become too narrow (AZ258, Etoposide)

Too large diameter

Neighbouring small cells can merge and small individual cells are lost (AZ-U, Lovastatin)

Why auto diameter

Cellpose estimated mean diameters from about 43 px to 96 px across compounds. One fixed scale cannot fit all phenotypes

Thresholds change which masks are accepted

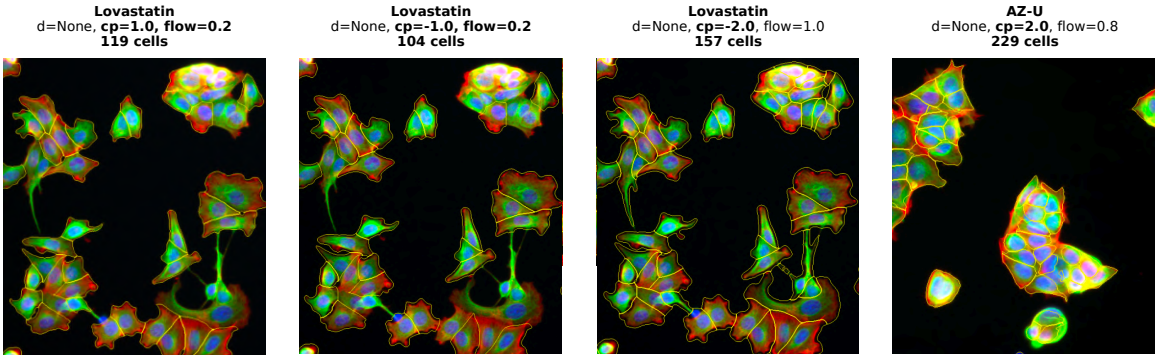


Figure 6. Cellpose threshold grid-search crops

Strict filter `flow=0.2`	Negative cell prob. cp=−1/−2	High cell probability cp=2
Complex or dim cells and whole clusters unsegmented	Too permissive, masks can include background or artifacts	Masks become too tight and can cut off real cell parts

Chosen Cellpose setting

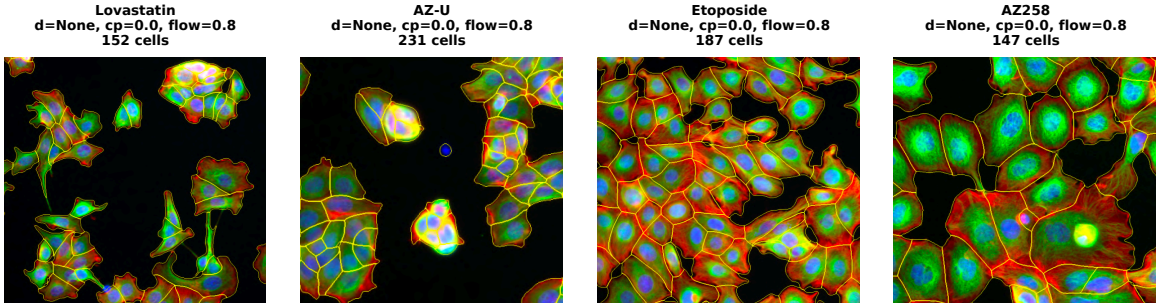


Figure 7. Selected Cellpose setting along compounds

Selection

Visually plausible masks were obtained with `cellprob_threshold=0.0` and 1, and with `flow_threshold=0.6` and 0.8.

Final configuration

`channels = actin + DAPI`, `diameter=None`
`cellprob_threshold=0.0`, `flow_threshold=0.8`

Segmentation quality control

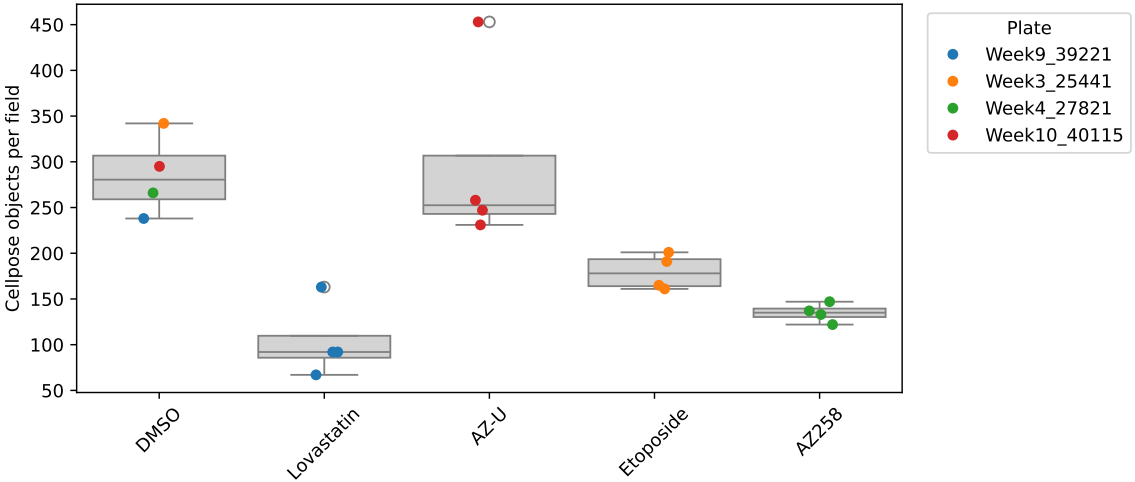


Figure 8. Cell counts per field by compound and plate

Morphometry table: 11 features, one row per cell

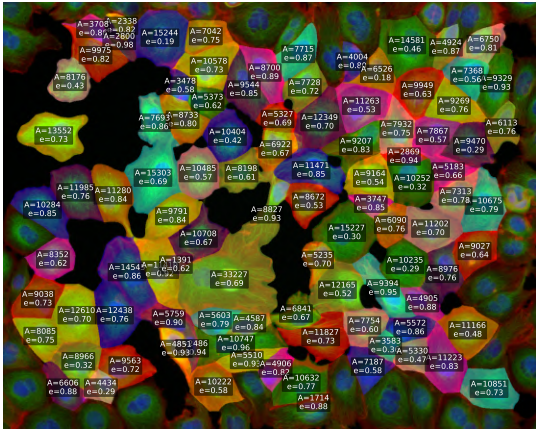


Figure 9. AZ258 morphometry example

- Features extracted with `skimage.measure.regionprops`
- Border-touching objects removed with `clear_border`
- Artifact filter:

$$300 \leq \texttt{area} \leq 25000$$

Table 2. Area distribution quantiles

Compound	0.01	0.05	0.50	0.95	0.99
DMSO	2.09	1328.65	3539.0	7011.10	8663.34
Lovastatin	429.15	923.90	2280.5	4130.85	6342.13
AZ-U	295.85	588.00	1452.0	3238.30	4059.16
Etoposide	1.00	2218.85	6009.5	12773.75	17691.27
AZ258	1.00	2331.40	9057.0	15312.85	17638.24

Violin plots for features

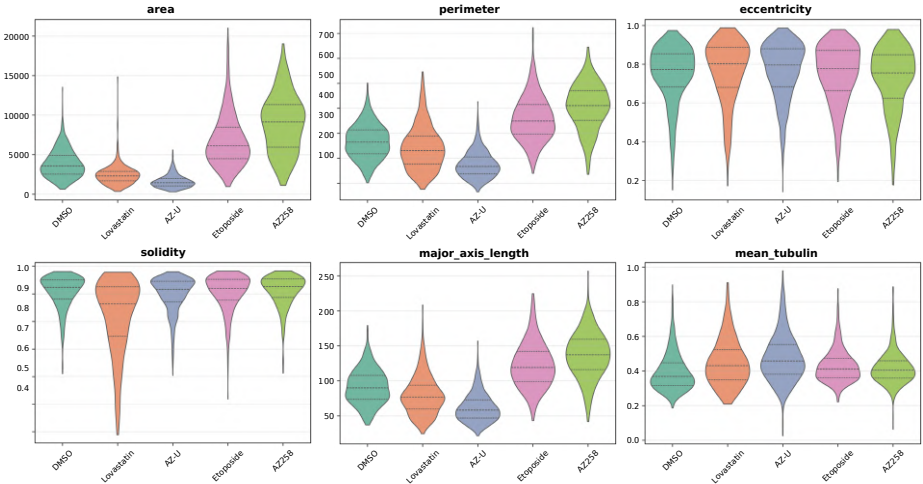


Figure 10. Distributions of some features across compounds

PCA shows class overlap

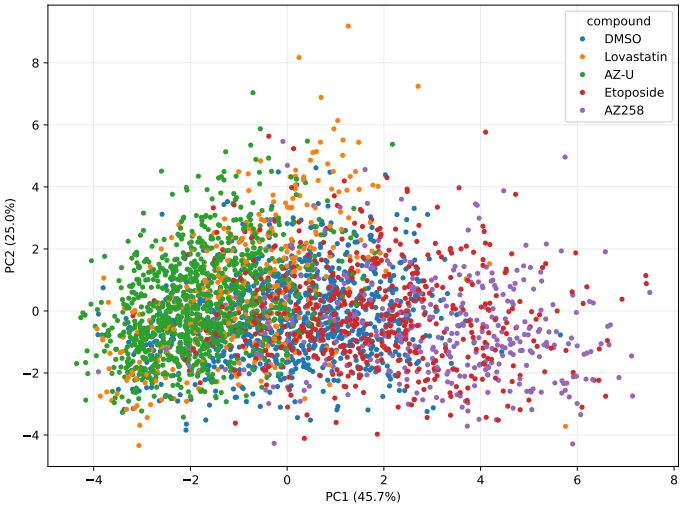


Figure 11. PC1/PC2 projection

Feature	PC1	PC2	PC3
minor_axis_length	**0.41**	-0.20	0.12
major_axis_length	**0.41**	0.17	0.14
area	**0.43**	-0.08	0.16
perimeter	**0.43**	0.08	0.15
equivalent_diameter	**0.44**	-0.06	0.13
mean_actin	-0.23	-0.05	**0.66**
mean_tubulin	-0.22	-0.04	**0.68**
extent	-0.02	**-0.49**	-0.06
aspect_ratio	-0.01	**0.52**	0.02
solidity	-0.05	**-0.40**	-0.08
eccentricity	-0.01	**0.50**	-0.01

Table 3. PCA feature loadings

Conclusion

PC1 is dominated by size features;
PC2 by shape and compactness;
PC3 by actin and tubulin intensity

Feature pairs

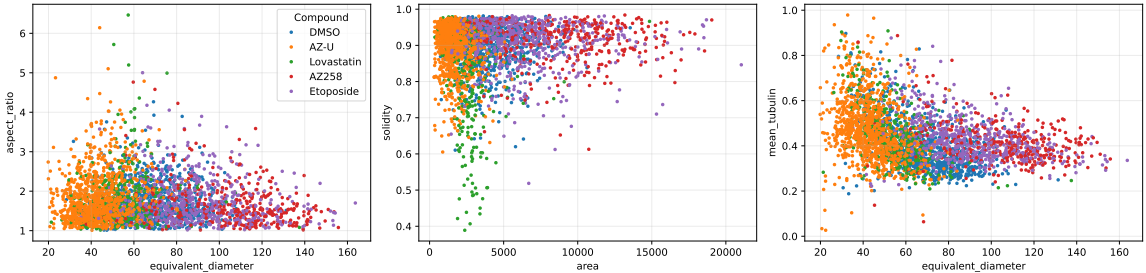


Figure 12. Morphometry pair plots

Note

For better class separation, mix features from different components: size + shape / intensity

Features above or below average

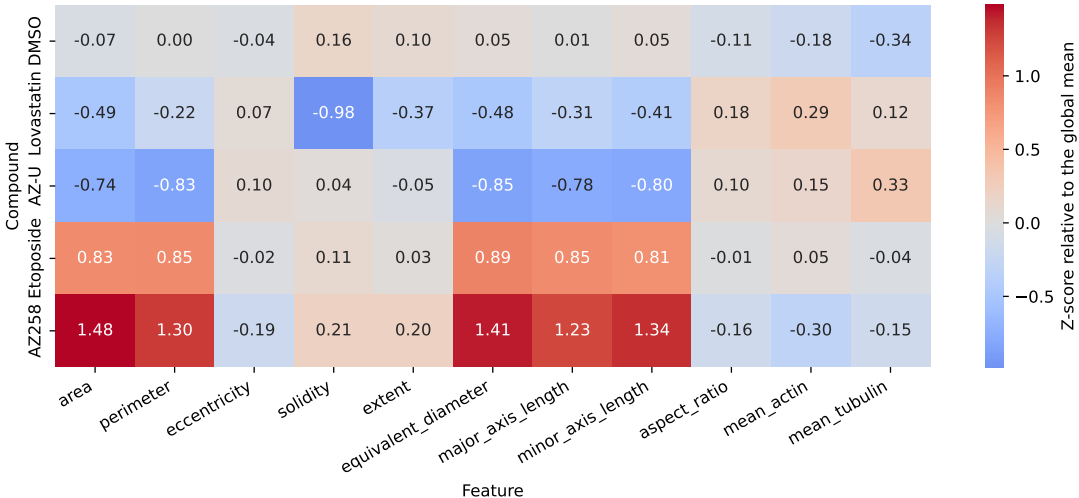


Figure 13. Z-score heatmap of morphometric features

Classification with and without DMSO control

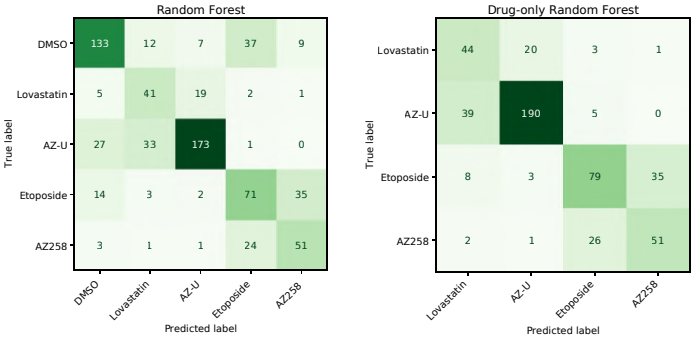


Figure 14. Confusion matrix

Table 4. Classifier performance

Model	Acc.	Macro F1	Wtd. F1
RF, drugs only	0.72	0.67	0.72
SVM, drugs only	0.70	0.65	0.71
RF	0.67	0.63	0.67
SVM	0.65	0.61	0.66
Log Reg	0.63	0.59	0.64

Table 5. F1-scores for classes

Model	DMSO	Lov.	AZ-U	Etop.	AZ258
RF, drugs only	–	0.55	0.85	0.66	0.61
SVM, drugs only	–	0.56	0.84	0.62	0.58
RF	0.70	0.52	0.79	0.55	0.58
SVM	0.69	0.49	0.81	0.52	0.56
Log Reg	0.67	0.47	0.78	0.50	0.52

Train/test split by `field_id`, stratified by compound: cells from one field cannot be in both train and test

Main limitations

- There is no ground-truth whole-cell annotation, so segmentation is evaluated visually by overlays and cell count checks.
- Only a small subset is used: 5 compounds, 4 middle-dose fields per compound.
- Cells from the same field, well or plate are not fully independent biological observations.
- 11 features describe cell size, shape and intensity, but do not capture detailed texture or cell-neighbourhood patterns.
- Dose-response analysis was started, but remains future work.

Main result

The project demonstrates a proof of concept. Morphometric features distinguish two treatment phenotypes, especially through size-related variables, but two compounds overlap.

Appendix: classifier hyperparameters

Shared setup

Split by field_id, stratified by compound: cells from one field cannot be in both train and test. class_weight="balanced" was used in LogReg, SVM, RF and Decision Tree to compensate for unequal class sizes. StandardScaler() was used for Logistic Regression and SVM, because they are scale-sensitive. random_state=SEED makes stochastic steps reproducible. Drug-only RF and SVM repeat the same setup after removing DMSO.

Logistic Regression

- max_iter=2000 optimization steps to converge.
- Defaults penalty=L2, C=1.0.
- L2 shrinks all coefficients smoothly; L1 push some coefficients to 0.
- C=1.0 is inverse regularization strength: smaller C stronger penalty, larger C weaker penalty.

SVM

- kernel="rbf", C=1.0, gamma="scale".
- C=1.0 controls the margin/error tradeoff: smaller C allows more errors and smoother boundary; larger C fits training data harder.
- gamma controls locality of the RBF kernel (how far the influence of an example): larger gamma - more local influence and complex boundaries, smaller gamma - broader influence and smoother boundaries.
- gamma="scale": sklearn sets gamma = 1 / (number of features x variance(X)) = 1/11.

Random Forest

- n_estimators=300, max_depth=8, min_samples_leaf=10.
- max_features="sqrt": each split tests only sqrt(number of features), about 3 of 11 features.
- bootstrap sampling is default.
- Why: nonlinear ensemble; depth and leaf size limit overfitting, while feature subsampling and bootstrap make trees less identical.

Decision Tree

- max_depth=4 (4-6 is ok).
- criterion="gini".
- Why: interpretable single-tree baseline; shallow depth keeps the split logic readable and limits overfitting.

Appendix: metrics

Confusion-matrix metrics

Accuracy	$(TP + TN)/(TP + TN + FP + FN)$
Precision$_c$	$TP_c/(TP_c + FP_c)$
Recall$_c$	$TP_c/(TP_c + FN_c)$
F1$_c$	$2P_cR_c/(P_c + R_c)$

Averaging

N_c = support, number of true test cells from class c.

Macro F1 $\quad \frac{1}{K}\sum_{c=1}^{K} \mathrm{F1}_c$

Weighted F1 $\quad \sum_{c=1}^{K} \frac{N_c}{N}\mathrm{F1}_c$

Macro gives all classes equal weight; weighted follows class frequency.

Z-score heatmap

$z_{c,f} = (\mu_{c,f} - \mu_f)/\mathrm{std}_f$. Mean in compound c minus global mean of feature f, divided by global std across all cells.

$\mathrm{var}_f = \frac{1}{n-1}\sum_{i=1}^{n}(x_{i,f} - \mu_f)^2$; $\mathrm{std}_f = \sqrt{\mathrm{var}_f}$. n - all cells.

Decision tree / Random Forest

Node impurity $\quad G(t) = 1 - \sum_{c=1}^{K} p(c|t)^2$

Split score $\quad G_{\mathrm{split}} = \frac{n_L}{n}G_L + \frac{n_R}{n}G_R$

$G = 0$: pure node; larger G: mixed classes. The tree chooses splits that reduce impurity. Random Forest averages votes from bootstrap-trained trees.

Jaccard index / IoU

For two masks A and B:

$$\mathrm{IoU}(A, B) = J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

1 means perfect overlap; 0 means no overlap.
Not used here: no manual whole-cell ground truth. Future use: Cellpose vs manual masks, or StarDist vs Cellpose nuclei masks.

Appendix: comparison of actin+tubulin merge, CLAHE and tuned Cellpose

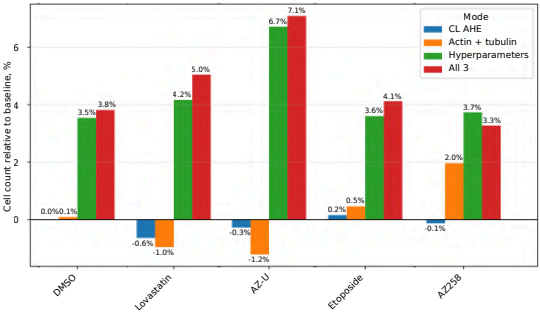


Figure 15. Cell counts across preprocessing modes

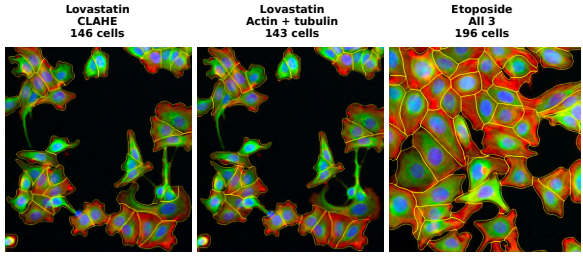


Figure 16. Cell crops in failed modes

Conclusion

The conservative tuned Cellpose configuration was kept for morphometry

Appendix: indirect nuclei count quality check with StarDist and Cellpose

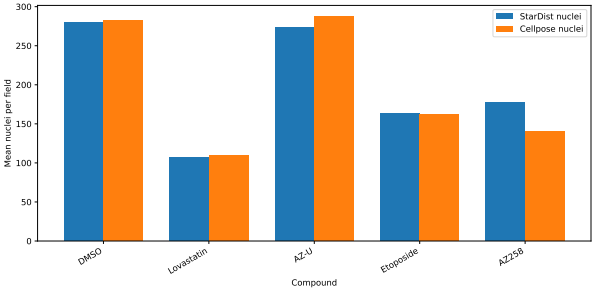


Figure 17. StarDist and Cellpose nuclei counts

AZ258, conc=0.1, well=C08
pink=StarDist, green=Cellpose, yel=both

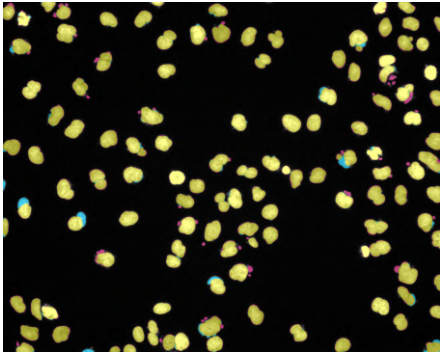


Figure 18. Nuclei overlap

Appendix: dose-response analysis

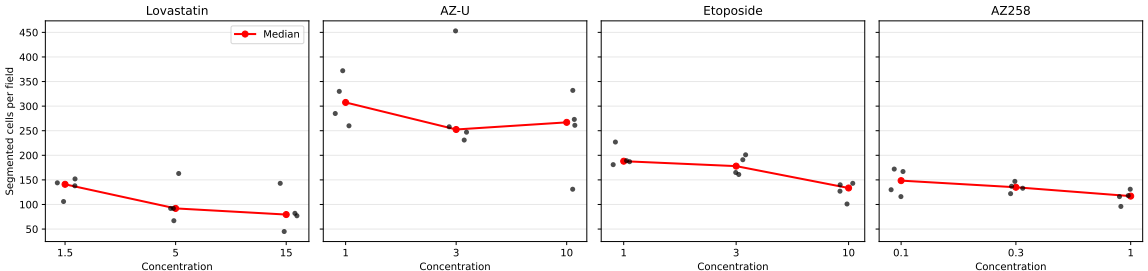


Figure 19. Dose response cell counts

Appendix: dose-response analysis

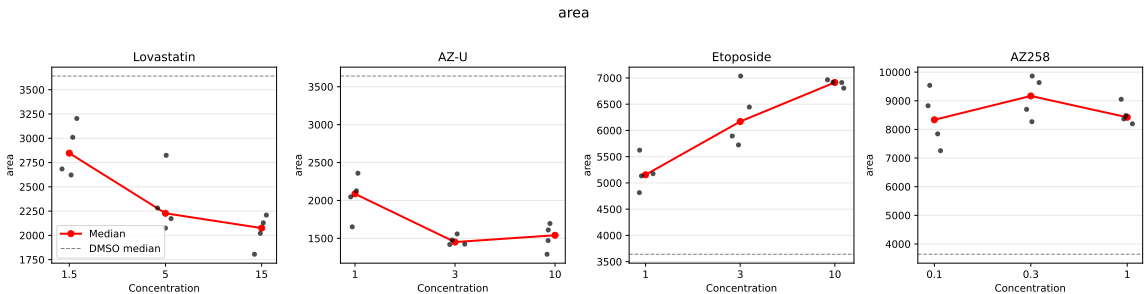


Figure 20. Dose response area

Appendix: dose-response analysis

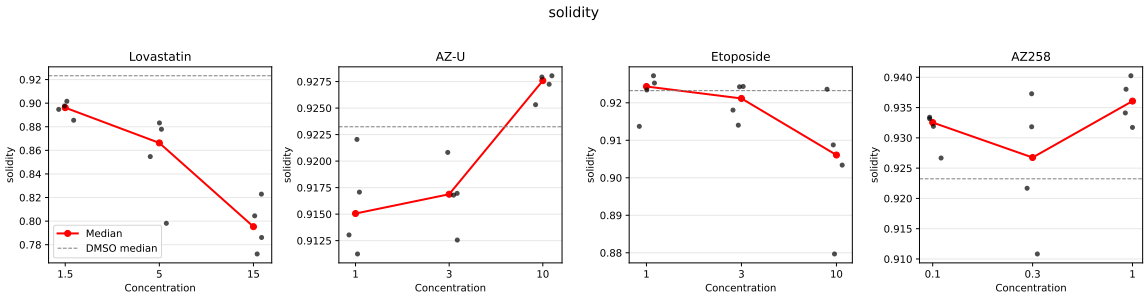


Figure 21. Dose response solidity

Appendix: dose-response analysis

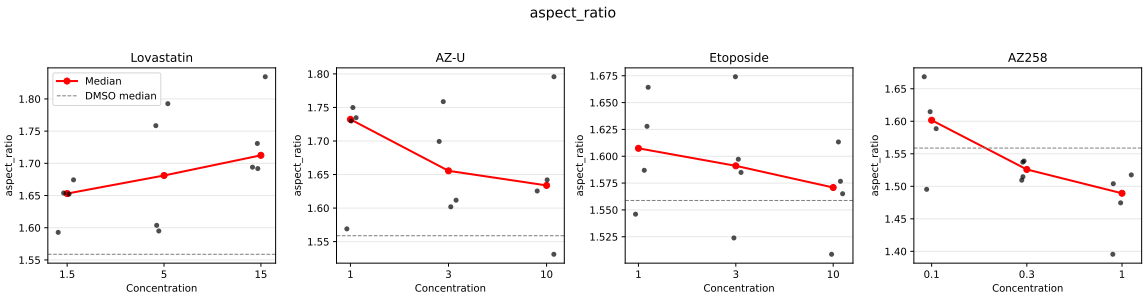


Figure 22. Dose response aspect ratio

Appendix: dose-response analysis

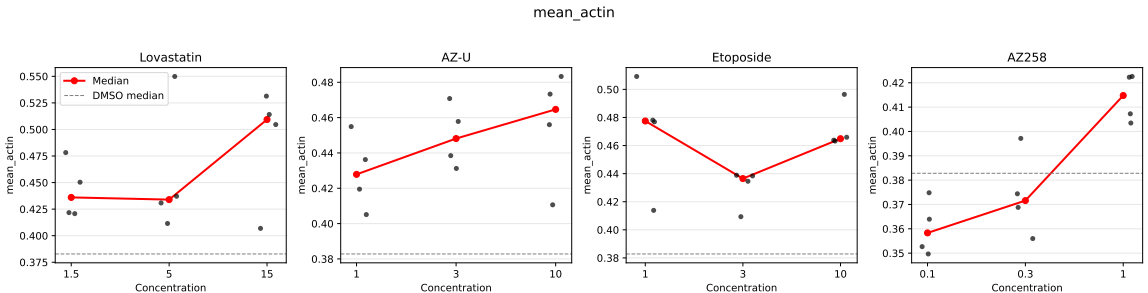


Figure 23. Dose response mean actin